

WROCLAW STRACHOWICE, Poland

WMO: 124240

Lat: 51.1032N

Lon: 16.8998E

Elev: 121

StdP: 99.88

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.5	-10.3	-16.2	0.9	-12.7	-13.3	1.2	-9.6	11.3	5.1	10.1	5.6	1.9	120	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.9	20.5	29.0	19.5	27.2	18.8	21.6	28.6	20.6	26.9	19.7	25.3	3.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.2	24.0	18.3	13.4	23.1	17.6	12.8	22.4	63.3	28.7	59.8	27.0	56.6	25.3	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.0	7.0	-16.5	34.1	3.9	1.9	-19.3	35.5	-21.6	36.6	-23.9	37.6	-26.7	39.0		
			-16.8	23.4	3.8	0.9	-19.6	24.0	-21.8	24.6	-24.0	25.1	-26.8	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	-0.5	1.0	4.3	9.6	14.1	17.6	19.6	19.2	14.6	9.8	4.9	1.0	(6)
		DBStd	8.15	5.28	4.94	4.20	4.03	3.62	3.48	3.11	3.08	3.32	3.97	4.02	4.77	(7)
		HDD10.0	1311	324	252	181	56	6	0	0	3	54	157	278		(8)
		HDD18.3	3331	583	485	436	263	138	55	21	26	119	266	402	536	(9)
		CDD10.0	1183	0	1	3	42	134	227	299	286	140	46	4	1	(10)
		CDD18.3	162	0	0	0	0	7	32	62	53	7	1	0	0	(11)
		CDH23.3	1706	0	0	0	17	104	340	637	521	84	3	0	0	(12)
		CDH26.7	504	0	0	0	1	17	96	218	154	18	0	0	0	(13)
(14)	Wind	WSAvg	3.1	3.6	3.6	3.6	3.2	3.0	2.9	2.7	2.6	2.8	3.0	3.3	3.6	(14)
(15)	Precipitation	PrecAvg	576	28	25	30	36	62	72	90	70	48	39	38	32	(15)
(16)		PrecMax	774	96	48	73	79	140	185	251	229	110	128	103	100	(16)
(17)		PrecMin	391	5	2	4	5	6	10	11	4	7	3	0	4	(17)
(18)		PrecStd	94	15	13	15	17	34	36	49	48	29	26	19	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	12.3	14.0	18.9	25.6	28.8	32.4	33.8	33.5	29.1	23.4	16.8	12.5		(19)
(20)		0.4% MCWB	9.0	9.4	11.3	15.8	18.7	21.7	20.8	21.4	18.9	16.4	12.5	9.7		(20)
(21)		2% DB	10.0	11.5	15.8	22.2	26.1	29.3	31.2	30.2	25.6	20.5	14.4	10.6		(21)
(22)		2% MCWB	7.5	8.2	9.8	13.9	17.5	20.0	20.2	20.4	17.4	15.1	11.3	8.2		(22)
(23)		5% DB	8.1	9.7	13.1	19.7	24.0	27.1	29.1	28.2	23.3	18.2	12.6	9.0		(23)
(24)		5% MCWB	6.0	7.0	8.8	12.6	16.6	18.7	19.5	19.4	16.6	14.1	10.1	7.1		(24)
(25)		10% DB	6.3	7.9	11.0	17.3	21.9	25.0	27.0	26.2	21.2	16.1	11.0	7.4		(25)
(26)		10% MCWB	4.6	5.7	7.7	11.5	15.2	17.6	19.0	18.6	15.7	12.9	8.9	5.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	9.5	10.1	12.4	16.6	19.7	22.8	22.7	22.5	19.9	17.4	13.2	10.1		(27)
(28)		0.4% MCDB	11.7	13.0	17.5	23.8	26.3	30.0	30.2	30.4	27.0	22.1	16.0	12.2		(28)
(29)		2% WB	7.7	8.5	10.5	14.9	18.4	21.0	21.5	21.4	18.5	15.6	11.6	8.4		(29)
(30)		2% MCDB	9.6	11.0	14.4	20.8	24.4	27.9	28.3	28.3	23.8	19.4	14.0	10.4		(30)
(31)		5% WB	6.1	7.2	9.2	13.2	17.2	19.6	20.6	20.3	17.4	14.5	10.3	7.2		(31)
(32)		5% MCDB	7.8	9.5	12.7	18.5	22.9	25.5	26.8	26.3	22.1	17.7	12.4	8.9		(32)
(33)		10% WB	4.6	5.7	7.8	11.9	16.0	18.4	19.7	19.4	16.3	13.2	9.0	5.8		(33)
(34)		10% MCDB	6.2	7.8	10.7	16.6	20.5	23.6	25.1	24.6	20.4	15.8	10.7	7.3		(34)
(35)	Mean Daily Temperature Range	MDBR	5.7	6.7	8.4	10.5	10.9	10.6	10.7	11.2	9.9	8.7	6.6	5.6		(35)
(36)		5% DB MCDBR	7.0	9.2	13.2	15.2	14.8	14.6	15.1	14.7	14.0	12.5	9.6	6.9		(36)
(37)		5% WB MCWBR	5.5	6.6	8.4	8.3	7.6	6.7	6.0	6.3	7.1	7.6	6.8	5.4		(37)
(38)		5% WB MCDBR	6.8	8.6	11.9	13.7	13.3	13.2	12.7	12.7	12.2	11.4	9.0	6.7		(38)
(39)		5% WB MCWBR	5.5	6.4	8.0	7.8	7.3	6.6	5.9	5.9	6.6	7.2	6.7	5.4		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.317	0.343	0.383	0.430	0.424	0.421	0.444	0.434	0.396	0.370	0.344	0.311		(40)
(41)		taud	2.321	2.299	2.223	2.141	2.205	2.246	2.195	2.234	2.311	2.346	2.366	2.360		(41)
(42)		Ebn at Noon	692	773	805	806	832	837	808	795	785	730	653	641		(42)
(43)		Edh at Noon	66	88	113	138	135	131	136	125	104	83	64	55		(43)
(44)	All-Sky Solar Radiation	RadAvg	0.78	1.48	2.57	4.13	5.07	5.34	5.04	4.50	3.21	1.84	0.89	0.58		(44)
(45)		RadStd	0.14	0.26	0.36	0.55	0.58	0.50	0.58	0.40	0.44	0.26	0.13	0.07		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.62	N/A	N/A	+0.91	+0.68	+0.70	N/A	-193	+118	+40	(46)
(47)	Regional (35 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+58	+20	(47)

Nomenclature: See separate page